

Data sheet

SonoMeter 40

Energy Meters

Description



The Danfoss SonoMeter 40 is a range of ultrasonic, compact energy meters intended for measuring energy consumption in heating and cooling applications for billing purposes. The meters are designed for remote meter read-out (AMR).

SonoMeter 40 energy meters consist of an ultrasonic flow sensor, a pair of Pt500 temperature sensors and a calculator with integrated circuits for temperature measurement, flow calculation and energy calculation.

Features

- Available in nominal flow rates q_p 0.6, 1.5, 2.5, 3.5, 6.0, 10.0, 15.0, 25.0, 40.0 and 60 m³/h
- Triple communication with M-Bus, wMBus onboard + 1 additional communication module slot
- Housings with thread (G3/4 to G2) or flange (DN 20 to DN 100) connections
- MID approval for ultrasonic energy meter with dynamic range of 1:100 ($q_i:q_p$) in class 2
- Optional MID approved dynamic range of 1:250 in class 2
- Temperature range 0 - 180 °C
- Nominal pressure PN25 bar as standard
- Calculator with IP65 protection class as standard (IP68 on special request)
- Flow sensor with IP68 protection class as standard
- Return or supply pipe installation, for vertical or horizontal mounting
- Battery lifetime for at least 15+1 years
- Low pressure loss, insensitive to dirt
- No calming sections needed before or after the flow sensor (DN 15-DN 50)
- For DN 65 to DN 100 calming sections are needed (5 x DN on inlet and 3 x DN on outlet of flow sensor)
- Remote reading via M-Bus, radio OMS 868 MHz, RS 485 Modbus, BACnet, LoRA-WAN, pulse output or optical interface
- Two pulse inputs or outputs (always included, optionally with pre installed cable)

Technical specifications

Complete energy meter

Application	Heating or Bifunctional (combined heating and cooling)	
Medium	Water quality with pH 7 to 9.5 VDI 2035, VdTÜV TCh 1466	
Measuring cycle	Flow: 1 sec. Energy: 1 sec. Temperature: 10 sec.	
Approved meteorological range	Openable version	Not openable version
	$\Delta\theta$: 3 - 175K	$\Delta\theta$: 3 - 110K
	θ : 0 - 180 °C	θ : 0 - 130 °C
	θ_q : 0.1 - 130 °C	θ_q : 0.1 - 130 °C
Temp. limits (energy calculation)	$\Delta\theta$: 0.2 - 180K	$\Delta\theta$: 0.2 - 110K
	θ : 0 - 180 °C	θ : 0 - 130 °C
Ambient operating temperature	Class C: 5 - 55 °C indoor installation, condensing, humidity max. 93%.	
Ambient storage temperature	temperature: -25 – +35 °C, humidity: max. 60%.	
Mechanical environment	Class M1	
Electromagnetic environment	Class E2	

Calculator

IP [EN60529]	65 (standard - enclosure openable) 68 (on special request - enclosure not openable)	
Battery Supply	Voltage	3.6V DC
	Battery type and size	2x Lithium Battery, AA-size
	Battery life time	15+1 years (including AMR communication)
Mains supply 24 V AC/DC (internal module)	Voltage	12 V - 36 V AC 12 V - 42 V DC Galvanically isolated
	Power consumption	not more than 20 mA
	Alternating current voltage frequency	50/60 Hz
	Backup supply	Internal backup battery: 1x AA-size Service life not less than 15 years (without data reading via a digital or wireless interface) - for powering of the meter, when the external power supply is turned off.
	Mains supply cable	2 wire copper cable with a diameter of $2 \times 0.25 \text{ mm}^2$
Mains supply 230 V AC (external transformer)	Voltage	230 VAC +10% -15% to 24VAC transformer for connection to mains power supply 24 AC/DC module.
	Power consumption	not more than 5 mA
	Alternating current voltage frequency	50/60 Hz
	Backup supply	Internal backup battery: 1x AA-size Service life not less than 15 years (without data reading via a digital or wireless interface) - for powering of the meter, when the external power supply is turned off.
	Mains supply cable	2 wire copper cable with a diameter of $2 \times 0.25 \text{ mm}^2$
Display	85 x 35 mm LCD display with 8 digits (11.5 mm high main characters) Menu guide and info panel	
Display Units	MWh - kWh - GJ - Gcal - °C - m ³ - m ³ /h	
Display Values	Energy - volume - flow rate - power - temperatures	
Optical interface	Always included	According to EN61107 (IEC1107)
Communication	To be specified when ordering the meter. Both options can be selected.	1. M-Bus according to EN13757-3, supports 300, 2400, 4800, 9600 baud. Delivered with 1,5 m cable. 2. wM-Bus 868 MHz (S1, T1 OMS)
Additional communication	Delivered from factory or later upgrade. One option from the list is possible.	1. Wired M-Bus 2. Modbus RTU 3. BACnet MS/TP 4. LoRaWAN

Technical specifications
(continuous)

Flow sensor

Diameter	DN 15	DN 20	DN 15	DN 20	DN 15	DN 20	DN 20	DN 25	DN 32	DN 25	DN 32	DN 40	DN 50	DN 65	DN 80	DN 100			
Nominal flow q _p (m³/h)	0.6		1.0		1.5		2.5		3.5		6		10	15	25	40	60		
Max flow q _s (m³/h)	1.2		2		3		5		7		12		20	30	50	80	120		
Minimum flow (1:100) q _l (m³/h)	0.006		0.01		0.015		0.025		0.035		0.06		0.1	0.15	0.35	0.4	0.6		
Cut off flow q _c (m³/h)	0.003		0.005		0.003	0.005		0.005		0.007		0.012		0.02	0.03	0.05	0.08	0.12	
Overflow q _{ss} (m³/h)	1.44		2.4		3.6		6		8.4		14.4		24	36	60	96	144		
Pressure drop at q _p (mbar)	70	9	113	25	171	58	72	198	94	90		100		180	120	200	180	180	
K _{vs} values	2.27	6.32	2.97	6.32	3.63	6.23	5.59	5.62	8.15	11.67		18.97		23.57	43.3	55.9	94.3	141.4	
Connection	G¾B	G1B	FL	G¾B	G1B	FL	G¾B	G1B	FL	G1¼B	FL	G1¼B	FL	G2B	FL	FL	FL	FL	
Overall length (mm)	110	190	110		190	110	165	190	130	130	190	260		260		300	270	300	360
Material of flow sensor	CW617N													CC770S					
Pressure PN (bar)	25 bar																		
IP [EN60529]	IP68																		
Temperature media (°C)	Θq: 0.1 - 130 °C																		
Cable length	1.2 m, 2.5 m or 5 m																		
Mounting	Any position; DN 15 - DN 50: No calming sections needed DN 65 - DN 100: calming sections are needed (5 x DN on inlet and 3 x DN on outlet)																		

Temperature sensor

Type	Pt500, 2 wire, according to EN1434, EN60751
Size / IP	DS type: Ø5.2 mm / IP65 PL type: Ø6 mm / IP67
Cable length	1.5 m, 3 m, 5 m or 10 m
Accuracy	Class B (EN60751)
Approved temp. range EN60751	Θ : 0 - 150 °C
Difference temp. EN60751	$\Delta\Theta$: 3 - 150 K
Pressure	25 bar
Mounting	without pockets for sizes G3/4 and G1 only

Communication interfaces
Optical interface

Optical interface is integrated into the front panel of the calculator. It is designed for data reading via M-bus protocol and parameterization of the meter using SonoMeter 40 UserConfig software and optical head OG-1-USB. The optical interface is activated by pressing the control button and shuts automatically off 5 minutes after the last pressing button or after completing data transmission via interface.

Wired M-Bus interface

- M-Bus protocol according to EN13757-3 standard
- 2 wire with polarity reversal protection
- Galvanic insulation
- Current drawn: one M-Bus load (1.52mA)
- Primary or secondary addressing supported
- Primary M-Bus address (M-Bus integrated) last 2 digits of a serial number; if "00" the primary address is 100.
- Primary M-Bus address (Additional M-Bus module) Address "2".
- Configurable baud rate 300/1200/2400/4800/9600, factory default 2400 baud rate.
- Battery lifetime min 15+1 years (2 x AA cell)
- Battery supply communication is limited on every 15 min at 2400 baud rate or faster, 70 energy meters on bus
- Fastest reading interval at mains supply: no limits
- Default M-Bus telegram includes below information's:
 - Date and time
 - Date and time of error starting
 - Error code
 - Battery operation time
 - Working time without error
 - Energy for heating
 - (Energy for cooling)
 - (Energy of tariff 1)
 - (Energy of tariff 2)
 - Volume
 - Pulse input 1
 - Pulse input 2
 - Power
 - Flow rate
 - Temperature 1
 - Temperature 2
 - Temperature difference
 - Serial number

Communication interfaces
(continuous)
Wireless M-Bus (OMS) interface, 868.95 MHz

- Wireless M-Bus protocol according to EN13757-4
- OMS (open metering system) compatible, compliant to OMS 4.0.2
- T1 mode (unidirectional)
- sending interval every 90 seconds (suitable for 'walk by' readings)
- battery lifetime min 15+1 years (2 x AA cell)
- default wireless M-Bus telegram includes below information's:
 - Energy for heating
 - Time
 - Volume
 - Energy for cooling
 - Pulse in 1
 - Pulse in 2
 - Power
 - Flow

Wireless M-Bus telegram parameter list and sending frequency is configurable!

Modbus RTU, RS485 module

Connectors	90 (noninverting, +) and 91 (inverting, -)
Baud rate	1200, 2400, 4800, 9600 (default), 38400, 56000, 57600, 115200 bps
Data format	8E1 (8 data bits, even parity bit, 1 stop bit) – default 8O1 (8 data bits, odd parity bit, 1 stop bit) 8N2 (8 data bits, no parity bit, 2 stop bits)
Power supply	Polarity independent connection for power supply – connectors 60 and 61 Voltage 12-24 V AC/DC Maximum power consumption 2 W max. Typical supply current 50 mA

BACnet MS/TP, RS485 module

BACnet MS/TP protocol and a serial interface for communication with external devices.

Connectors	90 (noninverting, +) and 91 (inverting, -)
Baud rate	9600, 19200, 38400 (default), 57600, 76800, 115200 bps
Data format	8E1 (8 data bits, even parity bit, 1 stop bit)
Power supply	Polarity independent connection for power supply – connectors 60 and 61 Voltage 12-24 V AC/DC Maximum power consumption 2 W max. Typical supply current 50 mA

LoRaWAN, 868 module

Transmitting scenarios:

- Heating: Energy and volume + 4 historical values.
- Heating and Cooling: Heating Energy, Cooling energy + Volume + 2 Historical values.

Pulse inputs

Number of pulse inputs	2 or no (to be specified when ordering)
indicated units	m ³
pulse value	programmable
input pulse types	IB according to LST EN1434-2
maximum permissible frequency	3 Hz
maximum permissible voltage	3.6 V
condition of maintenance of high level	3.6 V through 3.3 MΩ resistor

For versions with not openable enclosure (IP68), when meter is ordered with the pulse inputs, a permanently connected 1,5 m cable is fitted in the meter for connecting the pulse inputs.

Pulse outputs

Number of pulse outputs	2 or no (to be specified when ordering)
Class	OB in operating mode (OD in test mode)
Type	open collector
Permissible current	up to 20 mA
Voltage	up to 24 V
Pulse duration	125 ms in operating mode (1.2 ms in test mode)

Data sheet

SonoMeter 40

Communication interfaces (continuous)

Pulse value in the operating mode:

- when the output is configured for energy, the value of its pulses can be selected from the list

Energy measurement units	Possible values for the energy pulse *
„kWh“ or „MWh“	0,00001; 0,0001; 0,001; 0,01; 0,1; 1; 10; 100; 1000; 10000 MWh/pulse
„GJ“	0,0001; 0,001; 0,01; 0,1; 1; 10; 100; 1000; 10000 GJ/pulse
„Gcal“	0,0001; 0,001; 0,01; 0,1; 1; 10; 100; 1000 Gcal/pulse

* length of values list depends of permanent flow rate and LCD energy value comma position

- when the output is configured for water quantity, the value of its pulses can be selected from the list:
0,001; 0,01; 0,1; 1; 10 m³/pulse

For versions with not openable enclosure (IP68), when meter is ordered with the pulse outputs, a permanently connected 1,5 m cable is fitted in the meter for connecting the pulse outputs.

Data logger

Following hourly, daily and monthly parameter values are recorded in energy meter memory (can be configured by SonoMeter 40 UserConfig software):

- 1..... Integrated heating energy
- 2..... Integrated cooling energy
- 3..... Integrated energy of tariff 1
- 4..... Integrated energy of tariff 2
- 5..... Integrated volume of liquid
- 6..... Integrated pulse value in pulse input 1
- 7..... Integrated pulse value in pulse input 2
- 8..... Maximum thermal power value for heating and date
- 9..... Maximum thermal power value for cooling and date
- 10..... Maximum flow rate value and date
- 11..... Maximum value of flow temperature of heat conveying liquid and date
- 12..... Maximum value of return temperature of heat conveying liquid and date
- 13..... Minimum value of flow temperature of heat conveying liquid and date
- 14..... Minimum value of return temperature of heat conveying liquid and date
- 15..... Minimum value of temperature difference and date
- 16..... Average value of flow temperature of heat conveying liquid
- 17..... Average value of return temperature of heat conveying liquid
- 18..... Operating time without an error of thermal energy calculation
- 19..... Total error code
- 20..... Time when the flow rate exceeded 1.2 q_s
- 21..... Time when the flow rate was less than q_i

Data logger capacity

- up to 1480 h - for hourly records
- up to 1130 days - for daily records
- up to 36 last months - for monthly records
- Storage time of measured integrated parameters even if device is disconnected from power supply: not less than 15 years

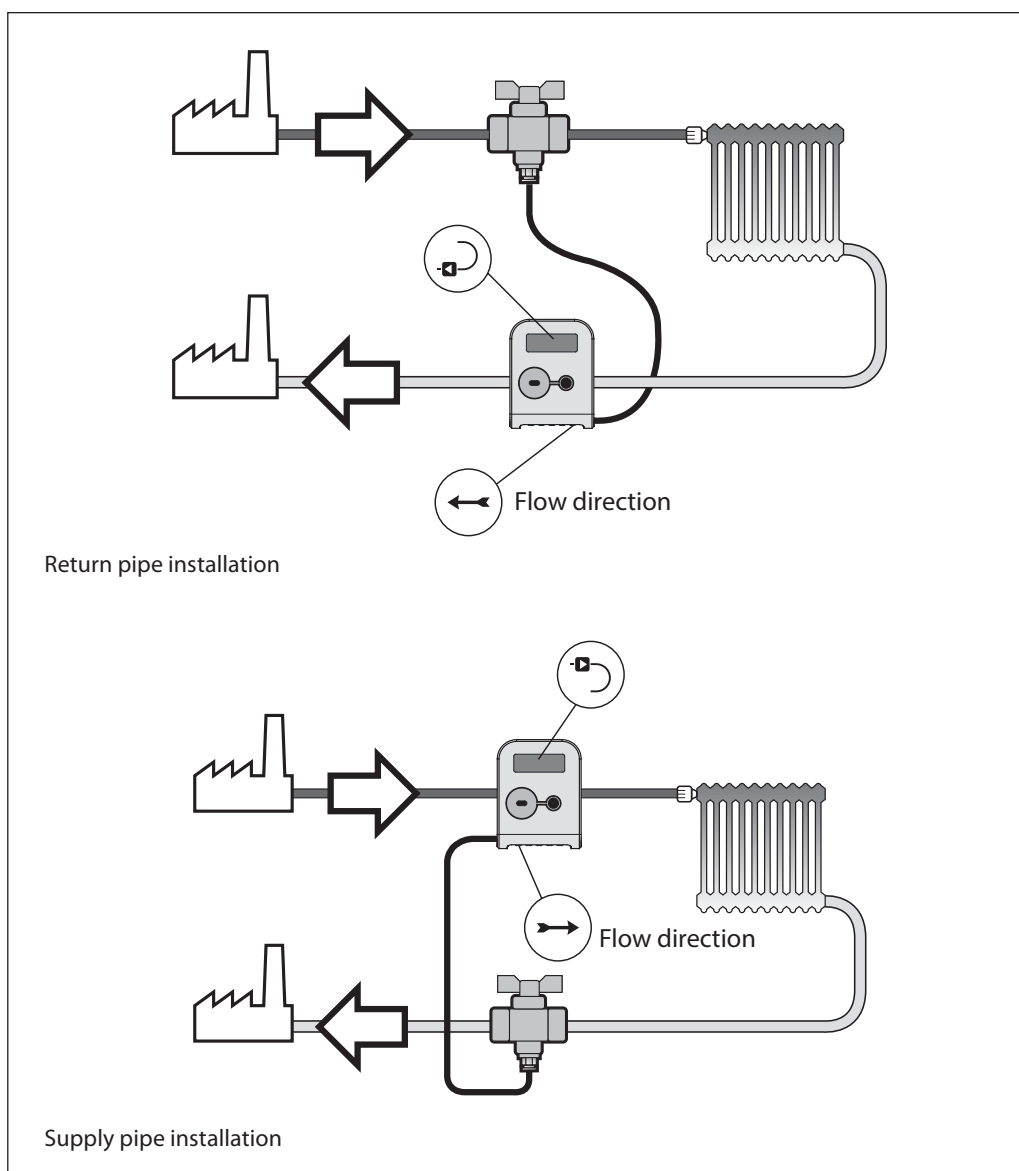
All data from archive can be read by means of the remote reading. In addition, data logger records of monthly parameters can be seen on the display.

Bifunctional meters

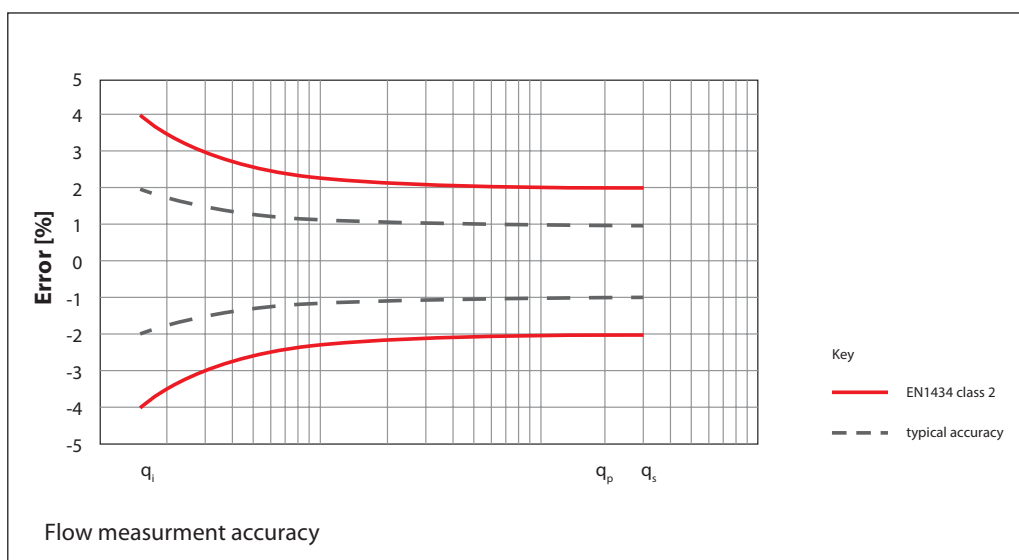
When the cooling energy tariff function is activated, in case of a negative temperature differential, energy will be accumulated in the additional tariff register Q*. In this case, energy values are calculated according to the following formulas:

- when the flow sensor is in the supply line
when $\Theta_1 > \Theta_2$: $Q = V \cdot \rho_1 \cdot (h_{T1} - h_{T2})$, $Q^* = 0$
when $\Theta_1 < \Theta_2$: $Q^* = V \cdot \rho_1 \cdot (h_{T2} - h_{T1})$, $Q = 0$
- when the flow sensor is in the return line
when $\Theta_1 > \Theta_2$: $Q = V \cdot \rho_2 \cdot (h_{T1} - h_{T2})$, $Q^* = 0$
when $\Theta_1 < \Theta_2$: $Q^* = V \cdot \rho_2 \cdot (h_{T2} - h_{T1})$, $Q = 0$

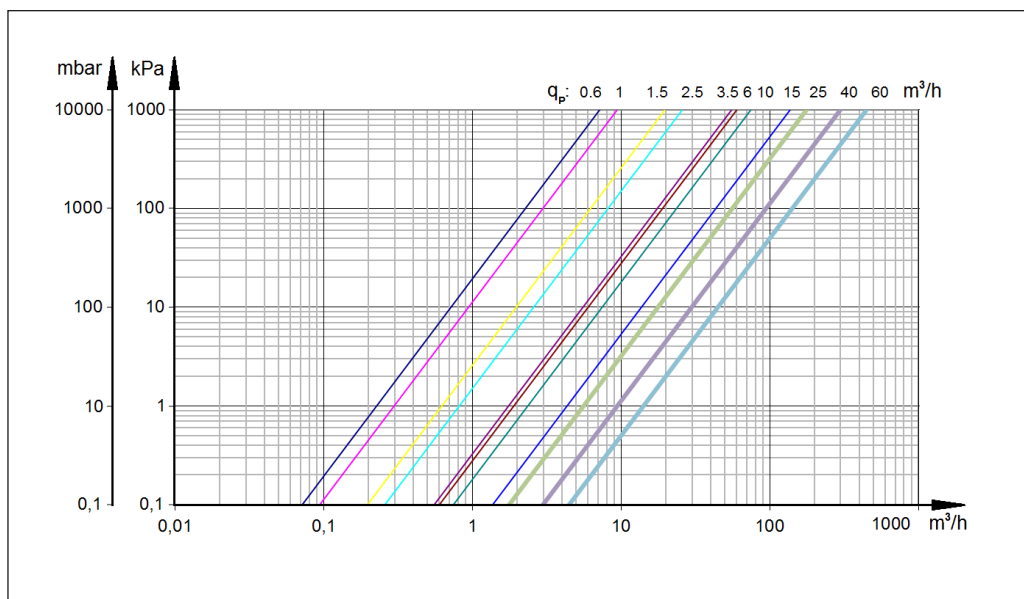
Application drawings



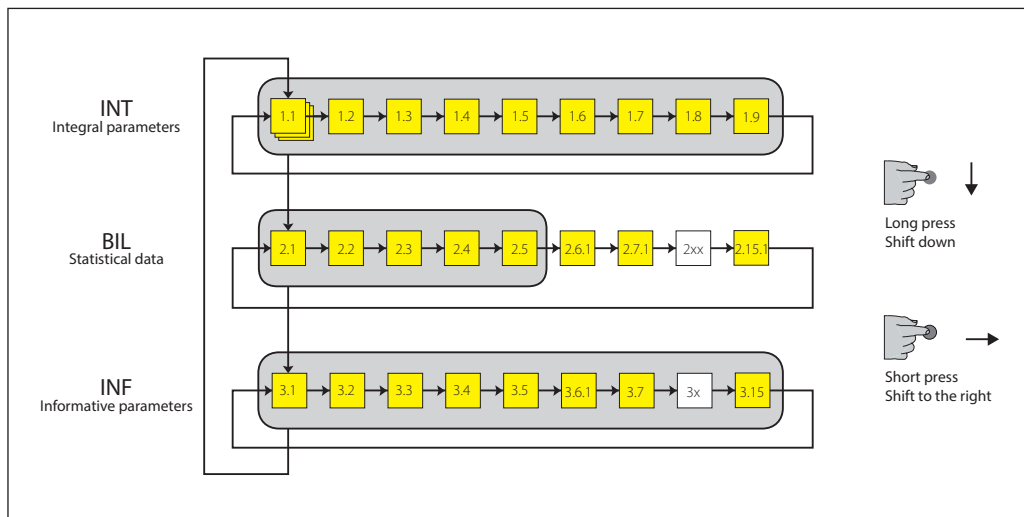
Accuracy



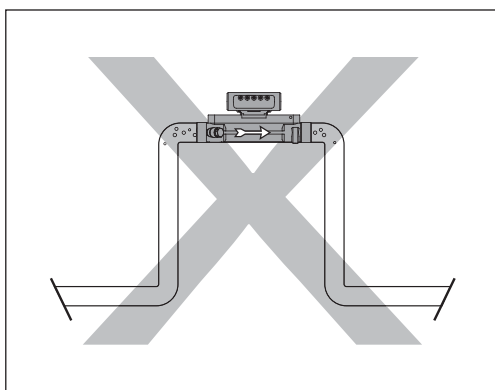
Pressure loss



Menu structure

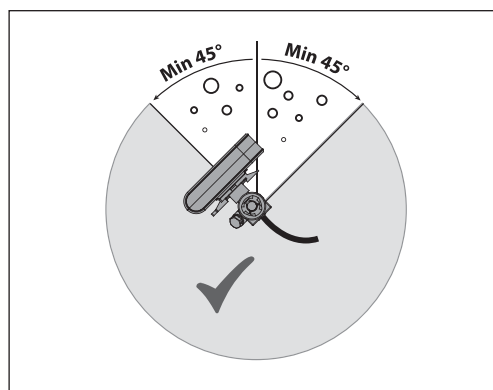


Mounting



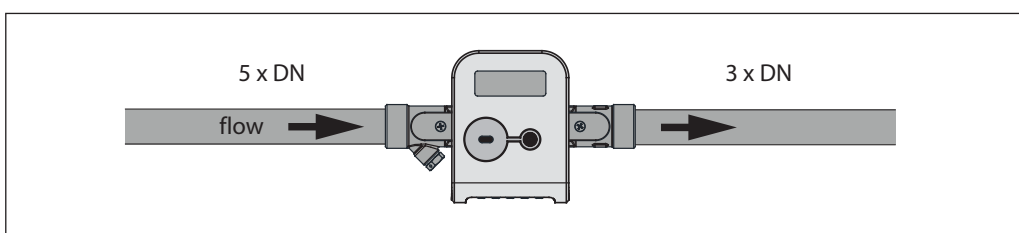
Pipe position:

No limitations but avoid positions where air can be collected.



Rotation in pipe axis:

Flow sensor should be angled in 45 to 315° to avoid air collection in flow sensor.

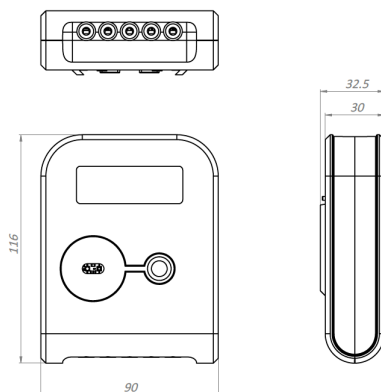


Inlet/outlet conditions (only for DN 65 - DN 100)

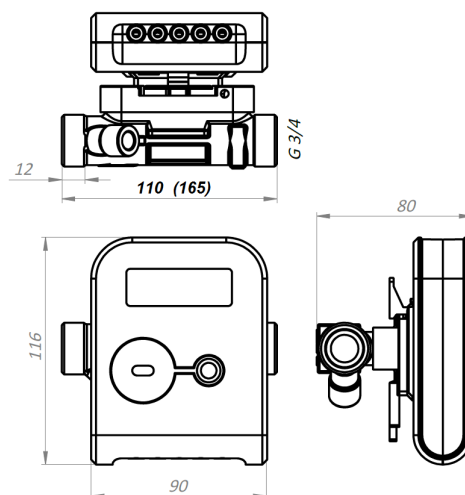
In order to maximize accuracy, it is necessary to have straight inlet and outlet flow conditions before and after the flow sensor: 5 × DN on inlet and 3 × DN on outlet of flow sensor.

Dimensions

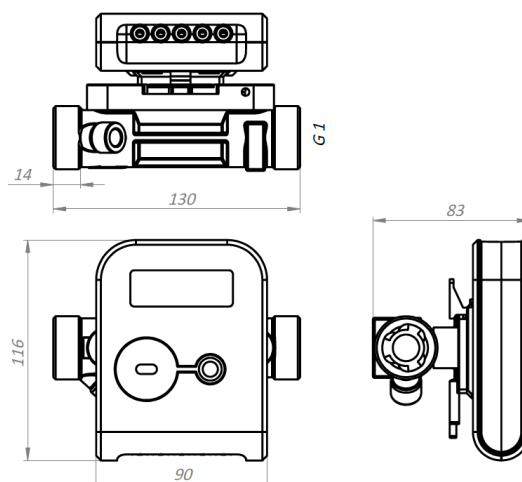
Calculator



Flow sensor qp= 0.6/1.0/1.5 m³/h
L=110 mm (L=165 mm);
connection type: thread G3/4"



Flow sensor qp= 1.5/2.5 m³/h
L=130 mm
connection type: thread G1"



Dimensions
(continuous)

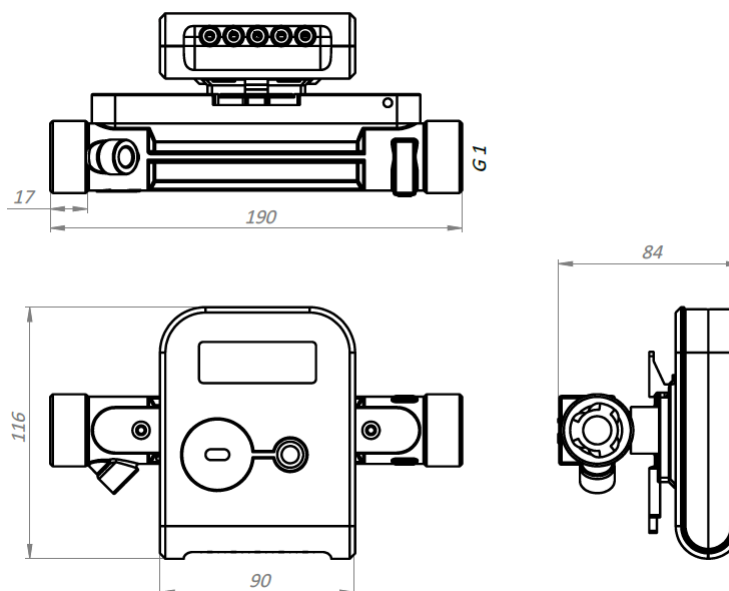
Flow sensor qp= 0.6/1.0/1.5/2.5 m³/h

L=190 mm

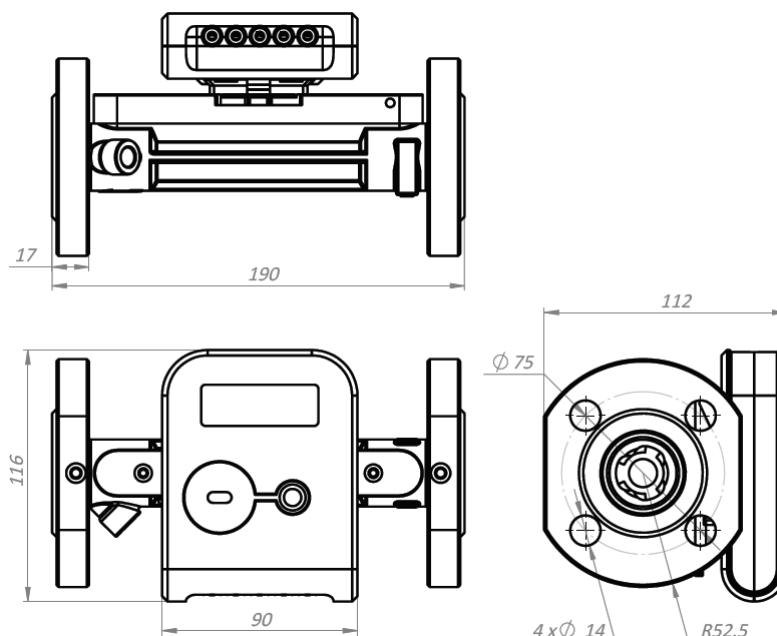
a) connection type: thread G1"

b) connection type: flanges DN20

a)



b)



Dimensions
(continuous)

Flow sensor qp= 3.5 m³/h

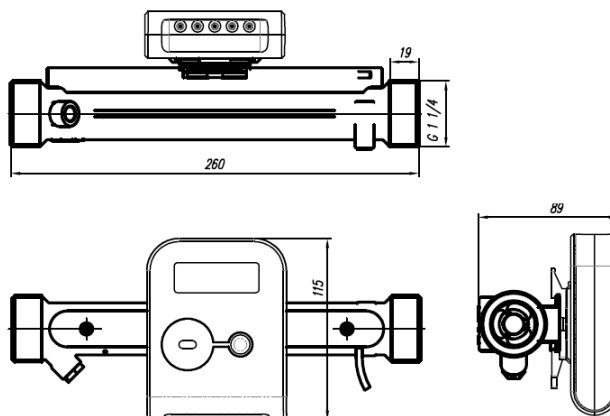
L=260 mm

a) connection type: thread G1 1/4"

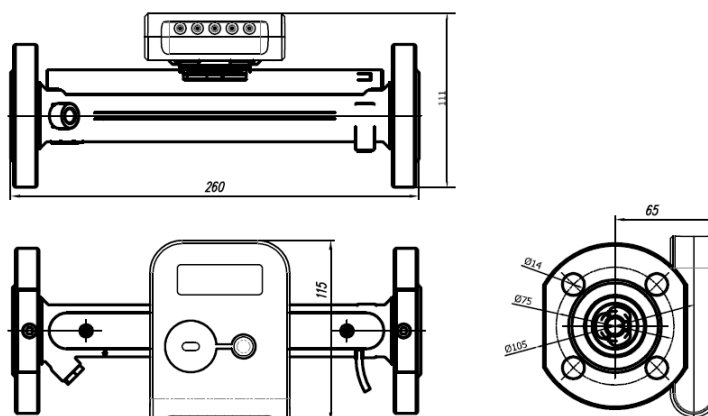
b) connection type: flanges DN25

c) connection type: flanges DN32

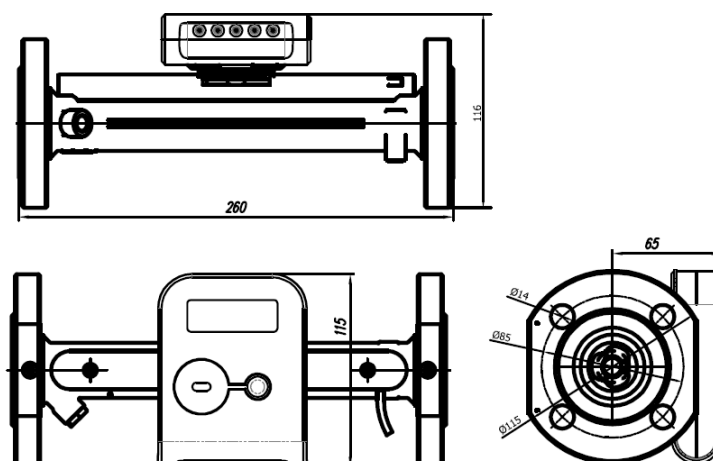
a)



b)



c)



Dimensions
(continuous)

Flow sensor qp= 6 m³/h

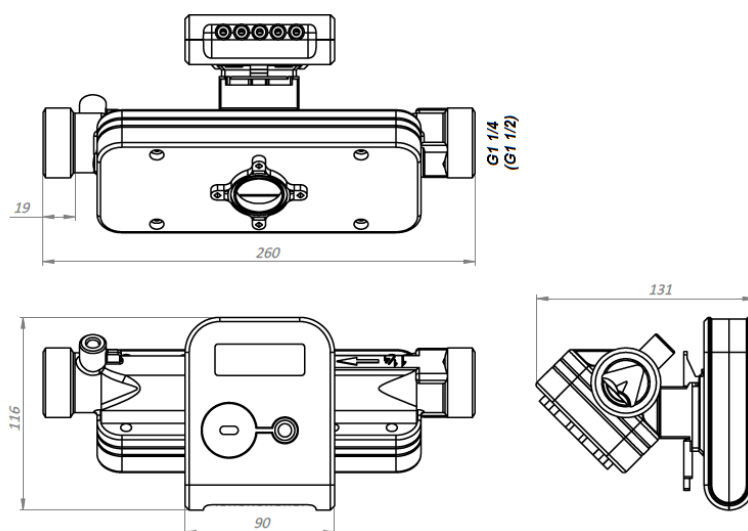
L=260 mm.

a) connection type: thread G1 1/4"

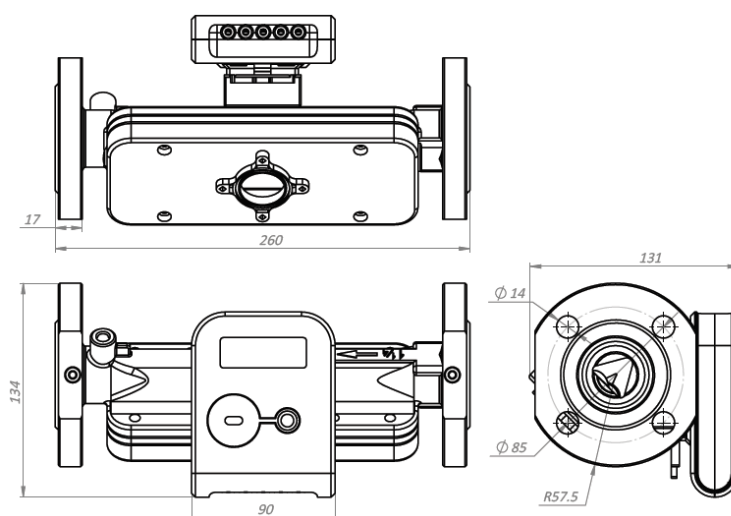
b) connection type: flanges DN25

c) connection type: flanges DN32

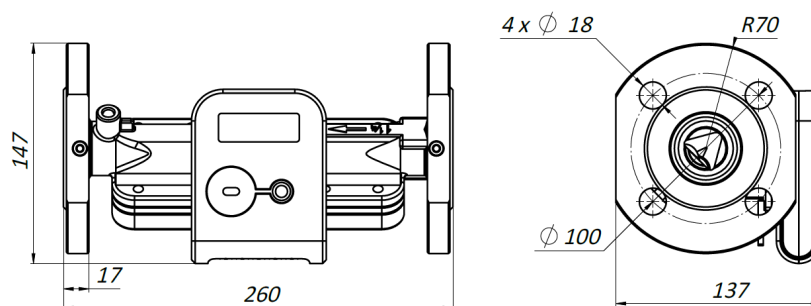
a)



b)



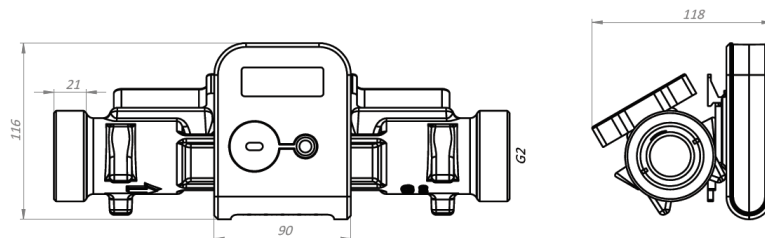
c)



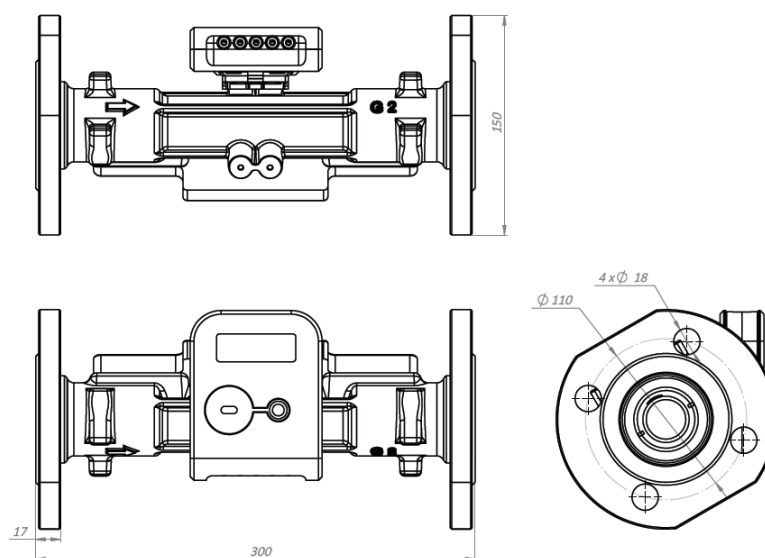
Dimensions
(continuous)

Flow sensor qp= 10.0 m³/h
L=300 mm
a) connection type: thread G2"
b) connection type: flanges DN40

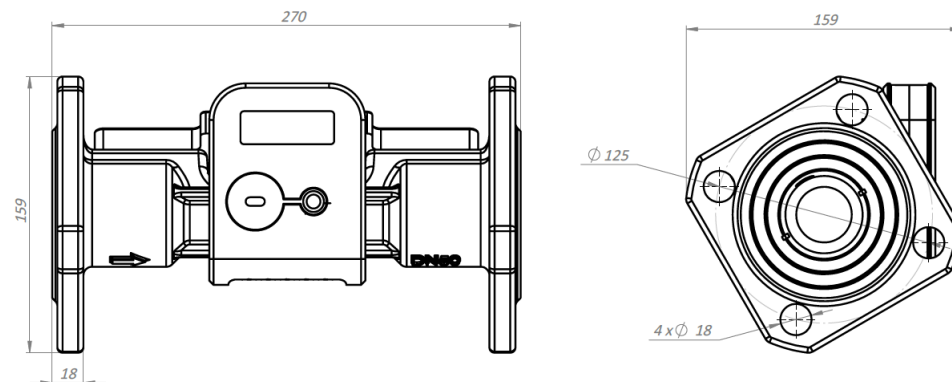
a)



b)

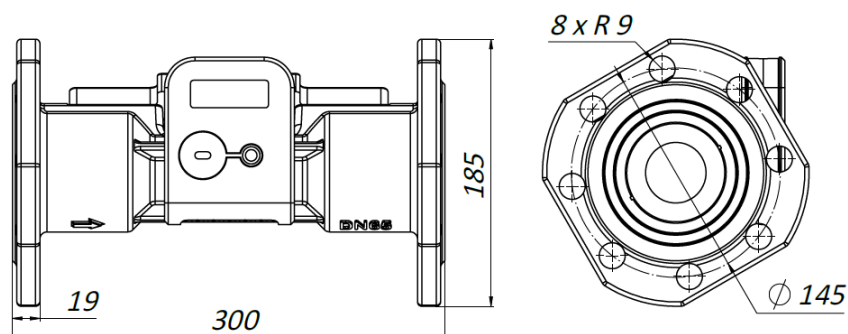


Flow sensor qp= 15 m³/h
L=270 mm
connection type: flanges DN50

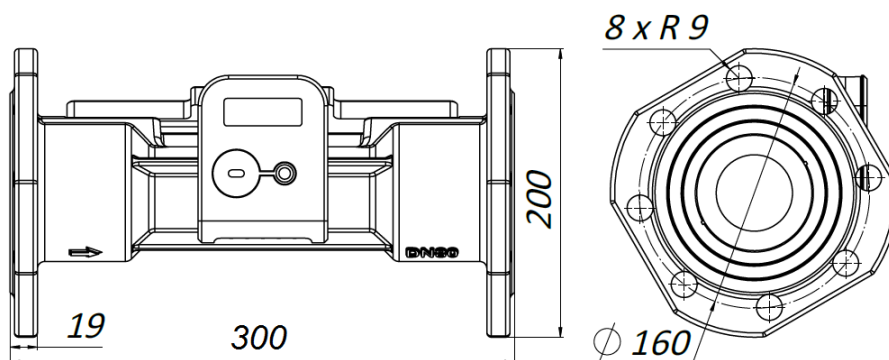


Dimensions
(continuous)

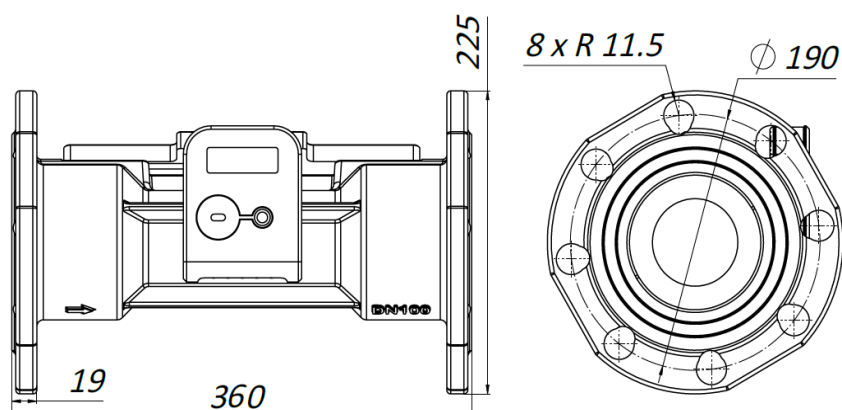
Flow sensor qp= 25 m³/h
L=300 mm
connection type: flanges DN65



Flow sensor qp= 40 m³/h
L=300 mm
connection type: flanges DN80



Flow sensor qp= 60 m³/h
L=360 mm
connection type: flanges DN100



Weight

Connection type (and length) of the flow sensor	Weight of the meter, maximum (kg)
G3/4" (110 mm)	0.7
G3/4" (165 mm)	0.8
G1" (110 mm)	0.7
G1" (130 mm)	0.8
G1" (190 mm)	0.9
DN20 (190 mm)	2.5
G1 ¼"	3.2
DN25	5.6
DN32	6.0
G2"	3.7
DN40	6.8
DN50	8.5
DN65	13
DN80	15
DN100	18

Ordering
SonoMeter 40 - Heating

Nominal flow, size and connection type	Installation	Cable length transducer cable	Cable length temperature sensor	Power supply	Communication integrated	Modules	Energy unit	Code no.
DN15 qp0.6m³/h 110mm G¾"	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2000
	Supply	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2001
	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	OMS	-	kWh	187F2026
	Supply	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	OMS	-	kWh	187F2027
	Return	1.2m	1.5m	Mains power 230V	MBus	-	kWh	187F2052
	Supply	1.2m	1.5m	Mains power 230V	MBus	-	kWh	187F2053
	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2084
DN15 qp1.5m³/h 110mm G¾"	Supply	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2085
	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2002
	Supply	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2003
	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	OMS	-	kWh	187F2028
	Supply	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	OMS	-	kWh	187F2029
	Return	1.2m	1.5m	Mains power 230V	MBus	-	kWh	187F2054
	Supply	1.2m	1.5m	Mains power 230V	MBus	-	kWh	187F2055
	Return	1.2m	1.5m	Mains power 230V	MBus	-	MWh	187F2078
	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2086
	Supply	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2087
DN20 qp2.5m³/h 130mm G1"	Supply	1.2m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2118
	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2006
	Supply	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2007
	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	OMS	-	kWh	187F2032
	Supply	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	OMS	-	kWh	187F2033
	Return	1.2m	1.5m	Mains power 230V	MBus	-	kWh	187F2058
	Supply	1.2m	1.5m	Mains power 230V	MBus	-	kWh	187F2059
	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2090
	Supply	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2091
DN20 qp2.5m³/h 190mm G1"	Supply	1.2m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2119
	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2008
	Supply	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2009
	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	OMS	-	kWh	187F2034
	Supply	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	OMS	-	kWh	187F2035
	Return	1.2m	1.5m	Mains power 230V	MBus	-	kWh	187F2060
	Supply	1.2m	1.5m	Mains power 230V	MBus	-	kWh	187F2061
	Return	1.2m	1.5m	Mains power 230V	MBus	-	MWh	187F2079
	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2092
DN20 qp2.5m³/h 190mm G1"	Supply	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2093

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Nominal flow, size and connection type	Installation	Cable length transducer cable	Cable length temperature sensor	Power supply	Communication integrated	Modules	Energy unit	Code no.
DN25 qp3.5m ³ /h 260mm G1¼"	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2010
	Supply	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2011
	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	OMS	-	kWh	187F2036
	Supply	2.5m	3m	Battery 3.6V DC (2 AA-cell)	OMS	-	kWh	187F2037
	Return	2.5m	3m	Mains power 230V	MBus	-	kWh	187F2062
	Supply	2.5m	3m	Mains power 230V	MBus	-	kWh	187F2063
	Return	1.2m	1.5m	Mains power 230V	MBus	-	MWh	187F2080
	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2094
	Supply	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2095
DN25 qp6.0m ³ /h 260mm G1¼"	Supply	1.2m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2120
	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2012
	Supply	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2013
	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	OMS	-	kWh	187F2038
	Supply	2.5m	3m	Battery 3.6V DC (2 AA-cell)	OMS	-	kWh	187F2039
	Return	2.5m	3m	Mains power 230V	MBus	-	kWh	187F2064
	Supply	2.5m	3m	Mains power 230V	MBus	-	kWh	187F2065
	Return	1.2m	1.5m	Mains power 230V	MBus	-	MWh	187F2081
	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2096
DN40 qp10m ³ /h 300mm G2"	Supply	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2097
	Supply	1.2m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2121
	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2014
	Supply	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2015
	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	OMS	-	kWh	187F2040
	Supply	2.5m	3m	Battery 3.6V DC (2 AA-cell)	OMS	-	kWh	187F2041
	Return	2.5m	3m	Mains power 230V	MBus	-	kWh	187F2066
	Supply	2.5m	3m	Mains power 230V	MBus	-	kWh	187F2067
	Return	2.5m	3m	Mains power 230V	MBus	-	MWh	187F2082
DN40 qp10m ³ /h 300mm threaded flange	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2098
	Supply	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2099
	Supply	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	MWh	187F2122
	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2016
	Supply	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2017
	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	OMS	-	kWh	187F2042
	Supply	2.5m	3m	Battery 3.6V DC (2 AA-cell)	OMS	-	kWh	187F2043
	Return	2.5m	3m	Mains power 230V	MBus	-	kWh	187F2068
	Supply	2.5m	3m	Mains power 230V	MBus	-	kWh	187F2069
DN50 qp15m ³ /h 270mm flange	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2100
	Supply	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2101
	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2018
	Supply	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2019
	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	OMS	-	kWh	187F2044
	Supply	2.5m	3m	Battery 3.6V DC (2 AA-cell)	OMS	-	kWh	187F2045
	Return	2.5m	3m	Mains power 230V	MBus	-	kWh	187F2070
	Supply	2.5m	3m	Mains power 230V	MBus	-	kWh	187F2071
	Return	2.5m	3m	Mains power 230V	MBus	-	MWh	187F2083
DN65 qp25m ³ /h 300mm flange	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2102
	Supply	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2103
	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2110
	Supply	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2111
	Supply	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	MWh	187F2123
	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	MWh	187F2020
	Supply	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	MWh	187F2021
	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	OMS	-	MWh	187F2046
	Supply	5m	5m	Battery 3.6V DC (2 AA-cell)	OMS	-	MWh	187F2047
DN65 qp25m ³ /h 300mm flange	Return	5m	5m	Mains power 230V	MBus	-	MWh	187F2072
	Supply	5m	5m	Mains power 230V	MBus	-	MWh	187F2073
	Return	5m	10m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2104
	Supply	5m	10m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2105
	Return	5m	10m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2112
	Supply	5m	10m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2113

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Nominal flow, size and connection type	Installation	Cable length transducer cable	Cable length temperature sensor	Power supply	Communication integrated	Modules	Energy unit	Code no.
DN80 qp40m ³ /h 300mm flange	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	MWh	187F2022
	Supply	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	MWh	187F2023
	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	OMS	-	MWh	187F2048
	Supply	5m	5m	Battery 3.6V DC (2 AA-cell)	OMS	-	MWh	187F2049
	Return	5m	5m	Mains power 230V	MBus	-	MWh	187F2074
	Supply	5m	5m	Mains power 230V	MBus	-	MWh	187F2075
	Return	5m	10m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2106
	Supply	5m	10m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2107
	Return	5m	10m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2114
DN100 qp60m ³ /h 360mm flange	Supply	5m	10m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2115
	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	MWh	187F2024
	Supply	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	MWh	187F2025
	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	OMS	-	MWh	187F2050
	Supply	5m	5m	Battery 3.6V DC (2 AA-cell)	OMS	-	MWh	187F2051
	Return	5m	5m	Mains power 230V	MBus	-	MWh	187F2076
	Supply	5m	5m	Mains power 230V	MBus	-	MWh	187F2077
	Return	5m	10m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2108
	Supply	5m	10m	Battery 3.6V DC (2 AA-cell)	MBus	-	GJ	187F2109
	Return	5m	10m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2116
	Supply	5m	10m	Battery 3.6V DC (2 AA-cell)	MBus	-	kWh	187F2117

Ordering

SonoMeter 40 - Combined heating and cooling

Nominal flow, size and connection type	Installation	Cable length transducer cable	Cable length temperature sensor	Power supply	Communication integrated	Modules	Energy unit	Code no.
DN15 qp0.6m ³ /h 110mm G¾"	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2600
	Supply	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2601
	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	OMS	-	-	187F2626
	Return	1.2m	1.5m	Mains power 230V	MBus	-	-	187F2639
	Return	1.2m	1.5m	Mains power 230V	OMS	-	-	187F2652
	Return	1.2m	1.5m	Mains power 230V	-	Modbus	-	187F2665
	Return	1.2m	1.5m	Mains power 24V AC/DC	-	Modbus	-	187F2678
	Supply	1.2m	1.5m	Mains power 230V	-	BACnet	-	187F2691
	Supply	1.2m	1.5m	Mains power 24V AC/DC	-	BACnet	-	187F2704
DN15 qp1.5m ³ /h 110mm G¾"	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2602
	Supply	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2603
	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	OMS	-	-	187F2627
	Return	1.2m	1.5m	Mains power 230V	MBus	-	-	187F2640
	Return	1.2m	1.5m	Mains power 230V	OMS	-	-	187F2653
	Return	1.2m	1.5m	Mains power 230V	-	Modbus	-	187F2666
	Return	1.2m	1.5m	Mains power 24V AC/DC	-	Modbus	-	187F2679
	Supply	1.2m	1.5m	Mains power 230V	-	BACnet	-	187F2692
	Supply	1.2m	1.5m	Mains power 24V AC/DC	-	BACnet	-	187F2705
DN20 qp1.5m ³ /h 130mm G1"	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2604
	Supply	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2605
	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	OMS	-	-	187F2628
	Return	1.2m	1.5m	Mains power 230V	MBus	-	-	187F2641
	Return	1.2m	1.5m	Mains power 230V	OMS	-	-	187F2654
	Return	1.2m	1.5m	Mains power 230V	-	Modbus	-	187F2667
	Return	1.2m	1.5m	Mains power 24V AC/DC	-	Modbus	-	187F2680
	Supply	1.2m	1.5m	Mains power 230V	-	BACnet	-	187F2693
	Supply	1.2m	1.5m	Mains power 24V AC/DC	-	BACnet	-	187F2706
DN20 qp2.5m ³ /h 130mm G1"	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2606
	Supply	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2607
	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	OMS	-	-	187F2629
	Return	1.2m	1.5m	Mains power 230V	MBus	-	-	187F2642
	Return	1.2m	1.5m	Mains power 230V	OMS	-	-	187F2655
	Return	1.2m	1.5m	Mains power 230V	-	Modbus	-	187F2668
	Return	1.2m	1.5m	Mains power 24V AC/DC	-	Modbus	-	187F2681
	Supply	1.2m	1.5m	Mains power 230V	-	BACnet	-	187F2694
		1.2m	1.5m	Mains power 24V AC/DC	-	BACnet	-	187F2707

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Nominal flow, size and connection type	Installation	Cable length transducer cable	Cable length temperature sensor	Power supply	Communication integrated	Modules	Energy unit	Code no.
DN20 qp2.5m ³ /h 190mm G1"	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2608
	Supply	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2609
	Return	1.2m	1.5m	Battery 3.6V DC (2 AA-cell)	OMS	-	-	187F2630
	Return	1.2m	1.5m	Mains power 230V	MBus	-	-	187F2643
	Return	1.2m	1.5m	Mains power 230V	OMS	-	-	187F2656
	Return	1.2m	1.5m	Mains power 230V	-	Modbus	-	187F2669
	Return	1.2m	1.5m	Mains power 24V AC/DC	-	Modbus	-	187F2682
	Supply	1.2m	1.5m	Mains power 230V	-	BACnet	-	187F2695
DN25 qp3.5m ³ /h 260mm G1¼"	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2610
	Supply	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2611
	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	OMS	-	-	187F2631
	Return	2.5m	3m	Mains power 230V	MBus	-	-	187F2644
	Return	2.5m	3m	Mains power 230V	OMS	-	-	187F2657
	Return	1.2m	1.5m	Mains power 230V	-	Modbus	-	187F2670
	Return	1.2m	1.5m	Mains power 24V AC/DC	-	Modbus	-	187F2683
	Supply	1.2m	1.5m	Mains power 230V	-	BACnet	-	187F2696
DN25 qp6.0m ³ /h 260mm G1¼"	Supply	1.2m	1.5m	Mains power 24V AC/DC	-	BACnet	-	187F2709
	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2612
	Supply	2.5m	3m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2613
	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	OMS	-	-	187F2632
	Return	2.5m	3m	Mains power 230V	MBus	-	-	187F2645
	Return	2.5m	3m	Mains power 230V	OMS	-	-	187F2658
	Return	1.2m	1.5m	Mains power 230V	-	Modbus	-	187F2671
	Return	1.2m	1.5m	Mains power 24V AC/DC	-	Modbus	-	187F2684
DN40 qp10m ³ /h 300mm G2"	Supply	1.2m	1.5m	Mains power 230V	-	BACnet	-	187F2697
	Supply	1.2m	1.5m	Mains power 24V AC/DC	-	BACnet	-	187F2710
	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2614
	Supply	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2615
	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	OMS	-	-	187F2633
	Return	2.5m	3m	Mains power 230V	MBus	-	-	187F2646
	Return	2.5m	3m	Mains power 230V	OMS	-	-	187F2659
	Return	2.5m	3m	Mains power 230V	-	Modbus	-	187F2672
DN40 qp10m ³ /h 300mm threaded flange	Return	2.5m	3m	Mains power 24V AC/DC	-	Modbus	-	187F2685
	Supply	2.5m	3m	Mains power 230V	-	BACnet	-	187F2698
	Supply	2.5m	3m	Mains power 24V AC/DC	-	BACnet	-	187F2711
	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2616
	Supply	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2617
	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	OMS	-	-	187F2634
	Return	2.5m	3m	Mains power 230V	MBus	-	-	187F2647
	Return	2.5m	3m	Mains power 230V	OMS	-	-	187F2660
DN50 qp15m ³ /h 270mm flange	Return	2.5m	3m	Mains power 230V	-	Modbus	-	187F2673
	Return	2.5m	3m	Mains power 24V AC/DC	-	Modbus	-	187F2686
	Supply	2.5m	3m	Mains power 230V	-	BACnet	-	187F2699
	Supply	2.5m	3m	Mains power 24V AC/DC	-	BACnet	-	187F2712
	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2618
	Supply	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2619
	Return	2.5m	3m	Battery 3.6V DC (2 AA-cell)	OMS	-	-	187F2635
	Return	2.5m	3m	Mains power 230V	MBus	-	-	187F2648
	Return	2.5m	3m	Mains power 230V	OMS	-	-	187F2661
	Return	2.5m	3m	Mains power 230V	-	Modbus	-	187F2674
	Return	2.5m	3m	Mains power 24V AC/DC	-	Modbus	-	187F2687
DN50 qp15m ³ /h 270mm flange	Supply	2.5m	3m	Mains power 230V	-	BACnet	-	187F2700
	Supply	2.5m	3m	Mains power 24V AC/DC	-	BACnet	-	187F2713
	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2717
		5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2718

Data sheet

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Nominal flow, size and connection type	Installation	Cable length transducer cable	Cable length temperature sensor	Power supply	Communication integrated	Modules	Energy unit	Code no.
DN65 qp25m ³ /h 300mm flange	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2620
	Supply	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2621
	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	OMS	-	-	187F2636
	Return	5m	5m	Mains power 230V	MBus	-	-	187F2649
	Return	5m	5m	Mains power 230V	OMS	-	-	187F2662
	Return	5m	5m	Mains power 230V	-	Modbus	-	187F2675
	Return	5m	5m	Mains power 24V AC/DC	-	Modbus	-	187F2688
	Supply	5m	5m	Mains power 230V	-	BACnet	-	187F2701
	Supply	5m	5m	Mains power 24V AC/DC	-	BACnet	-	187F2714
	Return	5m	10m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2719
DN80 qp40m ³ /h 300mm flange	Supply	5m	10m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2720
	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2622
	Supply	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2623
	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	OMS	-	-	187F2637
	Return	5m	5m	Mains power 230V	MBus	-	-	187F2650
	Return	5m	5m	Mains power 230V	OMS	-	-	187F2663
	Return	5m	5m	Mains power 230V	-	Modbus	-	187F2676
	Return	5m	5m	Mains power 24V AC/DC	-	Modbus	-	187F2689
	Supply	5m	5m	Mains power 230V	-	BACnet	-	187F2702
	Supply	5m	5m	Mains power 24V AC/DC	-	BACnet	-	187F2715
DN100 qp60m ³ /h 360mm flange	Return	5m	10m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2721
	Supply	5m	10m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2722
	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2624
	Supply	5m	5m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2625
	Return	5m	5m	Battery 3.6V DC (2 AA-cell)	OMS	-	-	187F2638
	Return	5m	5m	Mains power 230V	MBus	-	-	187F2651
	Return	5m	5m	Mains power 230V	OMS	-	-	187F2664
	Return	5m	5m	Mains power 230V	-	Modbus	-	187F2677
	Return	5m	5m	Mains power 24V AC/DC	-	Modbus	-	187F2690
	Supply	5m	5m	Mains power 230V	-	BACnet	-	187F2703
	Supply	5m	5m	Mains power 24V AC/DC	-	BACnet	-	187F2716
	Return	5m	10m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2723
		5m	10m	Battery 3.6V DC (2 AA-cell)	MBus	-	-	187F2724

Data sheet

SonoMeter 40

Ordering
Accessories

Product	Designation	Quantity	Code no.
Optical head	Optical head OG-1-USB	1 pc	187F3112
Power supply	Battery 3.6 V DC (AA cell)	1 pc	187F3113
	Mains unit 230 V AC (external module)	1 pc	187F3114
	Mains unit 24 V AC/DC	1 pc	-
Temperature sensor Ø 5.2 mm	Pt 500 / Ø 5.2 mm / 1.5 m cable, MID	1 pair	187F3125
	Pt 500 / Ø 5.2 mm / 3 m cable, MID	1 pair	187F3127
	Pt500 / Ø 5.2 mm / 5 m cable, MID	1 pair	187F3390
	Pt500 / Ø 5.2 mm / 10 m cable, MID	1 pair	187F3391
Temperature sensor Ø 6.0 mm	Pt500 / Ø 6.0 mm / 3 m cable, MID	1 pair	187F3123
	Pt500 / Ø 6.0 mm / 5 m cable, MID	1 pair	187F3124
	Pt500 / Ø 6.0 mm / 10 m cable, MID	1 pair	187F3389
Ball valve for Ø 5.2 mm temperature sensor Single pack	DN15 - 1/2" for direct sensor	1 pc	187F0593
	DN20 - 3/4" for direct sensor	1 pc	187F0592
	DN25 - 1" for direct sensor	1 pc	187F0591
	DN32 - 5/4" for direct sensor	1 pc	187F3448
Ball valve for Ø 5.2 mm temperature sensor Multi pack	DN15 - for direct sensor,12pc	12 pcs.	087H0118
	DN20 - for direct sensor,12pc	12 pcs.	087H0119
	DN25 - for direct sensor,12pc	12 pcs.	087H0120
	DN32 - for direct sensor,12pc	12 pcs.	187F3449
T-Fitting	DN15: G½", G½", G½" PN25 130 °C	1 pc	014U1959
	DN20: G¾", G½", G¾" PN25 130 °C	1 pc	014U1960
	DN25: G1", G½", G1" PN25 130 °C	1 pc	014U1961
	DN32: G1¼", G½", G1¼" PN25 130 °C	1 pc	014U1963
Pockets for Ø 5.2 mm temperature sensors	Ø 5.2 mm, brass, 35 mm length	1 pair	087G6053
	Ø 5.2 mm, brass, 52 mm length	1 pair	087G6054
	Ø 5.2 mm, brass, 85 mm length	1 pair	087G6055
	Ø 5.2 mm, brass,120 mm length	1 pair	087G6056
	Ø 5.2 mm, stainless steel, 85 mm length	1 pair	087G6057
	Ø 5.2 mm, stainless steel, 120 mm length	1 pair	087G6058
	Ø 5.2 mm, stainless steel, 155 mm length	1 pair	087G6059
	Ø 5.2 mm, stainless steel, 210 mm length	1 pair	087G6060
Pockets for Ø 6.0 mm temperature sensors	Ø 6.0 mm, brass, 40 mm length	1 pair	087G6061
	Ø 6.0 mm, brass, 85 mm length	1 pair	087G6062
	Ø 6.0 mm, brass, 120 mm length	1 pair	087G6063
	Ø 6.0 mm, stainless steel, 85 mm length	1 pair	087G6064
	Ø 6.0 mm, stainless steel, 120 mm length	1 pair	087G6065
	Ø 6.0 mm, stainless steel, 155 mm length	1 pair	087G6066
Tailpiece connection set	DN15 G3/4"-R1/2" PN25 130 °C	1 pair	087G6071
	DN20 G1"-R3/4" PN25 130 °C	1 pair	087G6072
	DN25 G1 1/4"-R1" PN25 130 °C	1 pair	087G6073
	DN40 G2"-R1 1/2" PN25 130 °C	1 pair	087G6074
Adapter for Ø 5.2 mm temperature sensor	R½" to M10x1 (1pc) - incl. copper gasket for flat sealing. PN25 130 °C	1 pc.	087G6075
	R½" to M10x1 (32pcs.) - incl. copper gasket for flat sealing. PN25 130 °C	32 pcs.	087G6076
	R½" to M10x1 (1pc) - conical thread for sealing with hemp. PN25 130 °C	1 pcs.	014U1941
	R½" to M10x1 (32pcs.) - conical thread for sealing with hemp. PN25 130 °C	32 pcs.	014U1935
Adapter for Ø 5.2 mm temperature sensor	M10x1 plastic installation set	20 pcs.	087G6077
	M10x1 brass installation set	20 pcs.	087G6078

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